Multimodal Video Summarization via Supervised Learning: Multimodal BiLSTM with Temporal Attention

and Evaluation of the VideoIntel Content Analysis Application

Tran Hai Duc
Student ID: 23127173
University of Science, VNU-HCM
thduc23@clc.fitus.edu.vn

Nguyen Cong Chien
Student ID: 23127331
University of Science, VNU-HCM
ncchien23@clc.fitus.edu.vn

***Abstract*—This report presents a supervised video summarization method evaluated on the SumMe and TVSum benchmarks. Visual features are extracted with a ResNet-50 backbone, while semantic speech features are obtained from Whisper automatic speech recognition and SentenceBERT sentence encodings. A two-layer Bidirectional LSTM (BiLSTM) combined with a Temporal Attention mechanism produces per-frame importance scores. Inference follows a fixed summary ratio of r=0.35 with Gaussian smoothing, shot segmentation, and boundary expansion to assemble the final dynamic summary. A complementary evaluation uses the VideoIntel application—built on CLIP, Whisper, and a local LLM—to measure the retention of search and question-answering capabilities when the original video is replaced by its summary. All content is self-contained; no external source files are required.**

***Index Terms*—video summarization, supervised learning, BiLSTM, temporal attention, multimodal, SumMe, TVSum, CLIP, Whisper, video question answering**

I. Introduction

A. The Video Summarization Problem

Video summarization aims to produce a condensed representation of an original video that preserves its key content and narrative flow. Two common output forms are distinguished:

- **Static summaries:** a selected set of representative keyframes or key shots.
- **Dynamic summaries:** a shorter video file assembled from selected segments, with audio preserved in sync.

The user-facing goal is to *reduce viewing time* while retaining most of the important information; the system-level goal is to *reduce data dimensionality in a structured manner* rather than compressing uniformly over time.

B. Methodological Context

Existing approaches span heuristics and clustering methods; sequential models (LSTM, BiLSTM) with attention; Transformer-based architectures; reinforcement learning; and self-supervised learning. **Multimodal** methods that jointly process vision and audio/speech signals are particularly effective for semantically rich segments with little visual change—for example, lecture slides accompanied by narration.

C. Contributions

The specific contributions of this report are:

1) A **multimodal supervised video summarization** pipeline trained on **SumMe** and **TVSum**, combining visual and speech features.
2) A **BiLSTM** architecture with **Temporal Attention** operating over 2432-dimensional multimodal feature sequences; the speech branch uses **Whisper** (ASR) and **SentenceBERT**.
3) A **dynamic video summary** V_{sum} with synchronized audio assembled via `ffmpeg` at a fixed summary ratio $r = 0.35$.
4) An auxiliary evaluation through the **VideoIntel** application, which measures **semantic search** (CLIP) and **question-answering** (Whisper + local LLM) quality on original/summary video pairs.

D. Related Work

General surveys [3]–[7] provide a broad overview of classical and deep-learning pipelines. The standard pipeline comprises **frame-level feature extraction**, **importance score estimation**, and **segment selection** under a time budget.

For supervised deep-learning methods, **LSTM/BiLSTM** variants such as vsLSTM and dppLSTM [11] model temporal dependencies; **attention** mechanisms (VASNet [16], Temporal Attention in this work) highlight informationally dense temporal regions. The foundational **Transformer** architecture [17] based on self-attention has inspired encoder-only and full encoder-decoder variants for video summarization [14], [18]. Unsupervised and adversarial directions (SUM-GAN [22], SUM-GAN-AED [24], Cycle-SUM [23]) and Transformer-based models (MSVA [20], PGL-SUM [19], MHSCNet [21]) provide additional architectural alternatives. **Graph Neural Networks** applied to spatial-temporal reasoning [26] capture object-level dependencies that frame-level methods miss. **Multimodal** fusion of vision and speech is an active research area [27], [28]; language-guided summarization via CLIP-based attention [29] shows strong zero-shot generalization.

At the feature level, **CNN ResNet** [31] provides visual vectors; the text branch uses **Whisper** [32] and **Sentence-BERT** [33] for fixed-dimension fusion. **CLIP** [34] provides a joint image-text embedding space for semantic search. Video recognition architectures (TimeSformer [35], Video Swin [36], I3D [37]) are relevant as downstream evaluation references. Comprehensive surveys cover deep learning methods for video summarization [3], [8]–[10].

II. Problem Formulation

A. Input

- An input video V in a standard format, of arbitrary duration.
- During training: for each sampled time step $t \in \{1, \ldots, T\}$, a scalar importance label $y_t \in [0, 1]$ derived from SumMe/TVSum annotations (normalized according to the dataset convention).

B. Output

- **Static summary (auxiliary):** a set of selected keyframes or shots.
- **Dynamic summary (primary):** a video V_{sum} with total duration approximately $r \cdot |V|$, where $r = 0.35$ is fixed at inference time. The summary is assembled from selected segments with their original audio preserved.

C. Deployment Constraints

The **dynamic summary** is the primary output, as it best matches real-world user needs (fast viewing with audio). Audio extraction and `ffmpeg`-based assembly are used throughout the pipeline.

III. Theoretical Background and Mathematical Formulation

A. Multimodal Feature Space

At each sampled time step t (e.g., at 2 fps), we construct a feature vector:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{v}_t \\ \mathbf{a}_t \end{bmatrix} \in \mathbb{R}^{D_v + D_a}, \quad D_v = 2048,\ D_a = 384,\ D_v + D_a = 2432. \tag{1}$$

Here $\mathbf{v}_t$ is the CNN visual feature (ResNet-50) and $\mathbf{a}_t$ is the temporally aligned semantic speech feature (Section V).

B. Supervised Learning and Binary Labels

The model produces a per-frame logit $z_t \in \mathbb{R}$. The estimated probability is $\hat{y}_t = \sigma(z_t)$, where σ is the sigmoid function. Ground-truth labels $y_t \in [0, 1]$ are **binarized** at threshold 0.5 to obtain $\tilde{y}_t \in \{0, 1\}$ for loss computation (BCEWithLogitsLoss). The **loss function** averaged over T steps is:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^{T} \Big[\tilde{y}_t \log \sigma(z_t) + (1 - \tilde{y}_t) \log\big(1 - \sigma(z_t)\big) \Big]. \tag{2}$$

C. Bidirectional LSTM (BiLSTM)

Given input sequence $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$, a forward LSTM produces $\overrightarrow{\mathbf{h}}_t \in \mathbb{R}^H$ and a backward LSTM produces $\overleftarrow{\mathbf{h}}_t \in \mathbb{R}^H$, with $H = 256$. The concatenated hidden state is:

$$\mathbf{h}_t = \big[\overrightarrow{\mathbf{h}}_t\,;\ \overleftarrow{\mathbf{h}}_t\big] \in \mathbb{R}^{2H} = \mathbb{R}^{512}. \tag{3}$$

Two stacked BiLSTM layers deepen the representation; each layer reads the sequence in **both temporal directions**, so the importance estimate at time t depends on both past and future context.

D. Temporal Attention

Let $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 512}$. Attention scores and weights are computed as:

$$e_t = \mathbf{u}^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}. \tag{4}$$

A globally pooled context vector is:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t\, \mathbf{h}_t. \tag{5}$$

To retain a **per-frame logit** for shot selection, $\mathbf{h}_t$ is combined with $\mathbf{c}$ via concatenation or a linear projection.

E. Output Logit Layer

$$z_t = \mathbf{w}_o^\top \mathbf{h}_t^{\text{out}} + b_o, \tag{6}$$

where $\mathbf{h}_t^{\text{out}}$ is the post-attention representation. Sigmoid is applied at inference as $\hat{y}_t = \sigma(z_t)$.

F. Datasets and Evaluation Metrics

- **SumMe** [1]: per-frame importance labels from multiple annotators.
- **TVSum** [2]: per-segment importance labels from multiple annotators.
- **F-score**: harmonic mean of precision and recall over selected vs. ground-truth frames/segments.
- **Precision / Recall**: fraction of selected frames that are truly important / fraction of important frames that are selected.
- **Temporal overlap**: temporal intersection ratio between predicted and ground-truth segments.

IV. Pipeline: Model Architecture and Forward Pass

A. Module Overview

The data-flow pipeline is as follows:

1) **Frame sampling** at 2 fps; sequence length capped at `max_seq_len` = 960 frames (≈8 min).
2) **Visual backbone:** ResNet-50 extracts $\mathbf{v}_t \in \mathbb{R}^{2048}$.
3) **Audio-language branch:** `ffmpeg` → Whisper → SentenceBERT → $\mathbf{a}_t$ (Section V).
4) **Feature concatenation:** $\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t] \in \mathbb{R}^{2432}$.
5) **Two-layer BiLSTM**, hidden size 256 per direction → $\mathbf{h}_t \in \mathbb{R}^{512}$.

6) **Temporal Attention** $\rightarrow$ linear head $\rightarrow z_t$.
7) Loss: **BCEWithLogitsLoss**; optimizer: **AdamW**; scheduler: **StepLR**.

B. Tensor Shapes and Batching

Sequences longer than 960 steps are **truncated**. Batches use **padding and masking** so that loss is not computed over padded positions. Each valid time step carries a 2432-dimensional input.

C. Ablation Variants

TABLE I: Ablation Variants and Architectural Differences

Variant	Modification
Full model	$\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t]$, with Temporal Attention
Visual-only	Drop $\mathbf{a}_t$; reduced input dimension
No Attention	Remove Temporal Attention; BiLSTM + linear head only
UniLSTM	`bidirectional=False`; single-direction LSTM

V. Pipeline: Audio Branch, Text Encoding, and Temporal Alignment

A. Audio Extraction

`ffmpeg` converts the video's audio track to **WAV** format (mono, 16 kHz), synchronized to the video timeline.

B. Automatic Speech Recognition (Whisper)

Whisper [32] produces **sentence/segment tokens** each carrying timestamps $(t_k^{\text{start}}, t_k^{\text{end}})$ and raw transcript text.

C. SentenceBERT Encoding

Each transcript segment is encoded into $\mathbf{s}_k \in \mathbb{R}^{384}$ using SentenceBERT [33].

D. Frame-Level Alignment

For each sampled frame at absolute time T_t:

- If $T_t \in [t_k^{\text{start}}, t_k^{\text{end}}]$, assign $\mathbf{a}_t = \mathbf{s}_k$.
- Otherwise: assign $\mathbf{a}_t = \mathbf{0}$ or interpolate from the nearest segment.

E. Feature Fusion

$$\mathbf{x}_t = [\mathbf{v}_t;\, \mathbf{a}_t] \in \mathbb{R}^{2432}. \tag{7}$$

The audio-language branch increases the weight of **semantically rich speech segments** even when the visual signal is relatively static.

VI. Pipeline: Training, Inference, and Video Assembly

A. Training (Offline)

1) **Pre-processing:** sample at 2 fps; ResNet-50 $\rightarrow$ visual tensors; Whisper + SentenceBERT $\rightarrow$ aligned audio-text features.
2) **Construct** $\mathbf{x}_t$ sequences; build padded/masked batches.
3) **Forward pass:** $\mathbf{x}_{1:T} \rightarrow$ BiLSTM $\rightarrow$ Attention $\rightarrow z_{1:T}$.
4) **Backward pass:** compute $\mathcal{L}$ (Eq. 2); update with AdamW; apply StepLR; early stopping; save `best.pt`.

B. Inference on a New Video

1) Extract $\{\mathbf{v}_t\}$ and $\{\mathbf{a}_t\}$ as in training.
2) Forward pass $\rightarrow \hat{y}_t = \sigma(z_t)$.
3) **Gaussian smoothing** to reduce temporal noise.
4) **Shot segmentation** via `keyshot.py`.
5) **Shot selection** such that total duration $\approx r \cdot |V|$, $r = 0.35$.
6) **Boundary expansion** of ± 1.5 s per selected segment; merge nearby segments if needed.
7) `ffmpeg` cuts and concatenates segments into V_{sum} with full audio.

VII. Hyperparameters and Experimental Configuration

TABLE II: Hyperparameters and Experimental Configuration

Component	Value / Notes
Architecture	Multimodal BiLSTM + Temporal Attention
Visual backbone	ResNet-50, $D_v = 2048$
Audio-text features	Whisper + SentenceBERT, $D_a = 384$
Input dimension	$2432 = 2048 + 384$
Frame sampling	2 fps, `max_seq_len` = 960
Smoothing	Gaussian, σ from config file
Summary ratio	$r = 0.35$
Boundary expansion	± 1.5 s
Loss	BCEWithLogitsLoss
Optimizer	AdamW, lr $= 10^{-3}$, wd $= 10^{-5}$
Scheduler	StepLR (step = 15, γ=0.5)

VIII. Experimental Results and Ablation Study

A. Ablation Study Design

An ablation study evaluates a model by *holding the dataset, train/test split, and evaluation protocol fixed* while *enabling or disabling individual architectural blocks*—for instance, dropping a feature branch, removing attention, or switching the LSTM directionality. Comparing metrics before and after each removal answers the question: *does this component carry useful signal, or does it merely add complexity?*

Four variants are examined, motivated as follows.

Full model (Multimodal BiLSTM + Temporal Attention). This is the complete proposed architecture: visual features from ResNet-50 are concatenated with SentenceBERT-encoded Whisper transcript embeddings, and a two-layer BiLSTM processes the joint sequence under a Temporal Attention mechanism.

Visual-only (no audio branch). Video summarization often relies primarily on visual content; however, for speech-rich material such as TVSum news clips, the language branch may supply contextual signal that is not recoverable from frames alone. Dropping $\mathbf{a}_t$ isolates the *marginal contribution of audio/text* relative to a vision-only CNN baseline [11].

No Temporal Attention. The attention module aggregates the global context vector $\mathbf{c}$, which enables the frame-level scorer to compare each position against the full sequence distribution. Removing attention reduces the architecture to a BiLSTM followed by a direct linear scorer, testing the *necessity of sequence-level context pooling* [16].

UniLSTM (unidirectional). A BiLSTM provides each position with both past and future context, which is particularly valuable for the importance-scoring task where a shot's relevance can only be judged relative to what comes later. Replacing `bidirectional=True` with `bidirectional=False` isolates the *gain from bidirectional encoding* [11], [19].

All four variants are trained with identical hyperparameters (Table II), the same random seed, and the same post-processing pipeline (r=0.35, Gaussian smoothing, boundary expansion ± 1.5 s).

B. Evaluation Protocol

Ground-truth labels are binarized using the *same top-k ratio* as the predicted summaries (controlled by `summary_ratio`). As a direct consequence, the number of positive predictions and the number of positive ground-truth frames are approximately equal, which causes **Precision and Recall to converge toward the same value** in the aggregate—this is a property of the evaluation protocol, not a numerical coincidence. F-score is therefore the harmonic mean of two nearly identical values and reduces to approximately Precision $\approx$ Recall for these runs. Temporal overlap is reported separately as a frame-level IoU between predicted and ground-truth segments, and is *not* affected by the top-k binarization symmetry.

SumMe and TVSum scores are kept separate throughout, as the two datasets differ substantially in video length, annotation style, and domain; merging them into a single number would obscure dataset-specific behavior.

C. Quantitative Ablation Results

TABLE III: Ablation Study: F-Score, Precision/Recall, and Temporal Overlap (%)

Variant	SumMe F (%)	TVSum F (%)	P/R (%)	Ovlp (%)
Full model (Ours)	26.9	60.1	47.7	23.5
Visual-only	31.4	59.7	49.1	33.3
No Attn.	34.9	63.7	52.9	27.4
UniLSTM	35.4	63.1	52.7	37.3

D. Analysis and Interpretation

Several observations merit discussion.

Dataset asymmetry. TVSum F-scores consistently exceed SumMe scores across all variants by a substantial margin ($\approx$27–28 pp). This reflects fundamental differences between the two benchmarks: TVSum videos are longer and their importance labels—derived from web-video titles—tend to be more concentrated, making top-k selection easier. SumMe uses crowdsourced per-frame scores with higher annotator disagreement, which is known to increase evaluation variance [1], [2]. Results on one benchmark should not be used to predict absolute performance on the other.

Full model does not rank first on SumMe. The full multimodal model achieves lower SumMe F-score than the visual-only, no-attention, and UniLSTM variants in this run. This outcome is consistent with known phenomena in multimodal learning: a richer architecture has more parameters and requires more careful regularization, learning-rate tuning, and early stopping; with a fixed training budget the simpler variants may converge to a better local optimum on a smaller dataset such as SumMe ($<$ 25 videos in the standard split) [19]. Additionally, if the audio transcripts in the SumMe subset are sparse or misaligned (SumMe contains user-generated short videos, many without sustained narration), the audio branch introduces noise rather than signal. A rigorous comparison would require multiple seeds with statistical significance testing; the present single-seed result should be interpreted as a *specific reproducible outcome* rather than a universal law.

Precision equals Recall in the aggregate Prec./Rec. column. As explained in Section VIII.B, the top-k binarization symmetry of the evaluator causes the aggregate positive counts of predictions and ground-truth to be nearly identical, so Precision $\approx$ Recall $\approx$ F-score in the combined metric column. The per-dataset SumMe and TVSum F-scores are computed from the respective subsets and are not subject to this equivalence.

Temporal overlap. The UniLSTM variant achieves the highest temporal overlap (37.3%), followed by visual-only (33.3%). The full model achieves the lowest overlap (23.5%). One interpretation is that the full model's attention mechanism concentrates high-score predictions in fewer, shorter temporal regions, yielding a sparser but more discriminative summary that overlaps less with the broad ground-truth intervals. An alternative explanation is that the audio branch shifts score peaks toward speech-dense regions that do not coincide with the annotated shot boundaries in this split. Both effects could be disentangled by examining score histograms and shot-length distributions, which are left for future work.

Implications for architecture design. The ablation results indicate that *architecture design choices interact with dataset characteristics* in non-trivial ways. The audio branch improves TVSum performance (60.1 vs. 59.7 for visual-only, a difference of +0.4 pp on a semantically richer set), while apparently hurting SumMe (-4.5 pp). Temporal Attention and bidirectionality follow a similar pattern. These findings suggest that for *mixed-content or narration-heavy* corpora, multimodal features are beneficial, whereas for *visually dynamic, low-speech* content, simpler unimodal models may be preferable. A dataset-adaptive weighting of modalities—for example, scaling $\mathbf{a}_t$ by an estimated speech-activity ratio—is a natural extension.

E. Comparison with Published Methods

Direct comparison should be interpreted cautiously due to differences in dataset splits, summary ratios, and post-processing strategies. The TVSum F-score of 60.1% is competitive with vsLSTM (54.2%) and approaches VASNet (61.4%), while SumMe performance remains below published baselines, consistent with the dataset-sensitivity discussion above.

TABLE IV: Comparison with Published Methods on SumMe and TVSum (F-Score, %)

Method	Year	Type	SumMe	TVSum
vsLSTM [11]	2016	Supervised	37.6	54.2
dppLSTM [11]	2016	Supervised	38.6	54.7
SUM-GAN [22]	2017	Unsupervised	38.7	50.8
Cycle-SUM [23]	2019	Unsupervised	41.9	57.6
VASNet [16]	2018	Supervised	49.6	61.4
DSNet [39]	2020	Supervised	51.2	61.9
MSVA [20]	2021	Supervised	53.4	61.5
PGL-SUM [19]	2021	Supervised	57.1	62.7
SUM-GDA	2020	—	56.3	60.7
This work (full model)	—	Supervised	**26.9**	**60.1**

F. Comparison with Initial Target Metrics

The initial targets were ≥ 0.43 F-score on SumMe and ≥ 0.55 on TVSum. The full-model TVSum result of **0.601** surpasses the TVSum threshold. The SumMe result of 0.269 falls below 0.43, reflecting the strong sensitivity of SumMe performance to dataset split, training budget, and audio-branch noise on short user-generated videos.

IX. VideoIntel Content Analysis Application

A. Motivation and Operating Hypotheses

The summarization pipeline produces a **shorter, lighter** file, but may:

1) **Reduce cost:** transcription time decreases with shorter audio.
2) **Reduce information:** loss of speech segments or off-topic selection $\rightarrow$ **semantic drift**.

VideoIntel measures both effects on **(original, summary) video pairs** using: indexing $\rightarrow$ Search $\rightarrow$ Q&A.

Technical assumptions: **CLIP** (ViT-B/32) provides a shared embedding space for text and frames; **Whisper** provides transcripts; the **local LLM (Ollama)** relies solely on the transcript—a short or off-topic transcript can cause **hallucination**.

B. Search Task

The scoring scheme is:

$$\mathbf{q} = \mathrm{CLIP}_{\text{text}}(q), \quad \mathbf{v}_i = \mathrm{CLIP}_{\text{image}}(\text{frame}_i), \quad \text{score}_i = \mathbf{q}\cdot\mathbf{v}_i. \tag{8}$$

Three fixed English queries benchmark CLIP top-1 score to estimate **visual representativeness**: "Person Speaking" (PS), "Important Event" (IE), and "Diagram or Chart" (DC).

C. Q&A Task

A fixed question is posed to every video: *"Please describe the overall content of this video based on the transcript. Answer in Vietnamese, approximately 3–5 sentences."* Generation is capped at 512 tokens for fair comparison.

D. Benchmark Metric Definitions

- $\text{duration_ratio} = T_{\text{sum}}/T_{\text{orig}}$
- File size reduction: $1 - \text{size}_{\text{sum}}/\text{size}_{\text{orig}}$
- Transcript coverage: $\sum \Delta t$ of Whisper segments as fraction of video duration
- Segment retention ratio: $N_{\text{seg,summ}}/N_{\text{seg,orig}}$
- CLIP top-1: cosine similarity between query and best-matching frame
- Whisper speedup: $t_{\text{orig}}/t_{\text{sum}}$

E. Quantitative Results (10 Video Pairs)

TABLE V: Average VideoIntel Benchmark Results (10 Video Pairs)

Metric	Value
Duration ratio (summary / original)	≈21.5%
File size reduction	≈76.8%
Whisper speedup	≈4.4×
Transcript coverage (original / summary)	≈93.8% / 94.2%
Segment retention ratio	≈25.6%
CLIP top-1 difference (summ − orig)	+0.0015 (negligible)
Acceptable Q&A (qualitative)	Original 8/10, Summary 5/10

TABLE VI: Duration and File Size Per Video Pair

Video	Orig (s)	Summ (s)	Ratio	Orig (MB)	Summ (MB)	Reduc.
video_01	351	76	21.7%	17.22	3.80	77.9%
video_02	1198	254	21.2%	75.29	15.90	78.9%
video_03	480	104	21.7%	25.86	4.75	81.6%
video_04	488	111	22.7%	21.48	8.76	59.2%
video_05	1555	342	22.0%	73.16	16.06	78.0%
video_06	927	193	20.8%	48.01	11.15	76.8%
video_07	1359	314	23.1%	71.41	16.05	77.5%
video_08	831	177	21.3%	49.93	10.43	79.1%
video_09	1054	210	19.9%	60.34	11.83	80.4%
video_10	1006	209	20.8%	54.61	11.77	78.4%
Avg.	825	179	21.5%	50.6	11.1	76.8%

TABLE VII: Whisper Transcription and CLIP Encoding Times (60 Keyframes)

Video	W-Orig (s)	W-Summ (s)	Speedup	C-Orig (s)	C-Summ (s)
video_01	109.29	50.52	2.2×	3.03	3.15
video_02	112.15	25.20	4.5×	3.45	3.32
video_03	97.64	15.91	6.1×	3.46	3.44
video_04	207.15	29.73	7.0×	3.16	3.09
video_05	132.18	36.23	3.6×	3.20	3.12
video_06	155.04	44.66	3.5×	3.31	3.21
video_07	194.76	44.11	4.4×	3.12	3.32
video_08	117.31	25.34	4.6×	3.07	2.99
video_09	148.50	33.30	4.5×	3.26	3.34
video_10	155.26	38.14	4.1×	3.13	3.13
Avg.	142.9	34.3	4.4×	3.22	3.21

TABLE VIII: Transcript Coverage and Segment Retention

Video	Cov Orig	Cov Summ	Seg Orig	Seg Summ	% Ret.
video_01	98.4%	100.0%	103	31	30.1%
video_02	93.7%	99.4%	478	71	14.9%
video_03	93.1%	93.0%	89	19	21.3%
video_04	91.5%	97.7%	84	42	50.0%
video_05	96.1%	93.5%	320	138	43.1%
video_06	94.1%	92.3%	202	51	25.2%
video_07	88.7%	94.0%	400	81	20.3%
video_08	93.4%	85.3%	164	33	20.1%
video_09	92.5%	89.7%	378	36	9.5%
video_10	96.8%	97.2%	214	46	21.5%
Avg.	93.8%	94.2%	243	55	25.6%

TABLE IX: Average CLIP Top-1 Scores: Original vs. Summary (Three Queries)

Query	Original	Summary	Diff.
Person Speaking (PS)	0.2509	0.2524	+0.0015
Important Event (IE)	0.2354	0.2374	+0.0020
Diagram or Chart (DC)	0.2274	0.2271	−0.0003
Mean	0.2379	0.2390	+0.0015

The overall mean CLIP top-1 difference (summary − original) is **+0.0015**, which is **statistically negligible**.

F. Qualitative Q&A Results

Ratings: **Accurate** (topic matches, no factually incorrect details), **Partial** (correct direction but missing details or hallucinated extras), **Incorrect** (off-topic or wrong content).

Overall: On the original transcript, **8/10** responses were acceptable. On the summary transcript, only **5/10** were Accurate; the remaining 5 fell into Partial or Incorrect.

TABLE X: Per-Video Qualitative Q&A Rating (Summary Transcript)

Video	Rating	Observation
video_01	Incorrect	Original: US–Israel–Iran news. Summary: completely different topic (antibiotics / comedy) — severe content-selection error.
video_02	Accurate	Both cover Israel/Iran, energy, US social issues — topic matches.
video_03	Partial	Correct sports direction (FA Cup) but missing specific team/player details.
video_04	Accurate	Both cover AI scenario planning and control risks — well aligned.
video_05	Accurate	BCIs, biomaterials in summary matches original neural electronics content.
video_06	Partial	Correct geopolitical direction but adds "Vietnamese traffic accident" absent from original (hallucination).
video_07	Partial	Correct on economy/China but adds "Vietnam economic competition" which is unrelated.
video_08	Partial	Correct on Iran/Trump/China but adds off-topic Chinese real estate and New York protests.
video_09	Partial	International part approximately correct; domestic content missing (only ≈9.5% of segments retained).
video_10	Accurate	Both cover Middle East maritime issues — key topics preserved.

Interpretation. Q&A quality on the summary does *not* track CLIP search quality: transcript density and topic fidelity are the critical factors. The two extremes are **video_01** (complete topic drift) and **video_09** (9.5% segment retention causing severe information loss).

G. Limitations and Recommendations

- Sample size: 10 videos; one LLM; one question template.
- CLIP top-1 score does not substitute for human-labeled relevance.
- **Recommendations:** (i) enforce a minimum segment count before trusting Q&A; (ii) check topic consistency between transcripts; (iii) verify audio coverage on anomalous drops.

X. Architecture and Training Optimization

This section describes four targeted modifications to the baseline architecture of Section IV, applied jointly to form an *optimized variant* that is evaluated on the same SumMe/TVSum test split. Each modification addresses a specific, identifiable bottleneck in the baseline, drawing on established theoretical principles and prior work in video summarization, multi-objective training, and deep-learning optimization. The motivation for each change is stated explicitly before its mathematical formulation, and the interaction between modifications is discussed at the end.

A. Evaluation Protocol and Baseline Recap

Both the baseline and the optimized checkpoints are evaluated on the **same held-out test split**, with an identical random seed, summary ratio r=0.35, and post-processing pipeline (Gaussian smoothing, boundary expansion ± 1.5 s, and `ffmpeg`-based assembly). The four reported metrics are: **Precision**, **Recall**, **F-score** (all on binarized top-k frame labels), and **temporal overlap** (frame-level intersection-over-union between predicted and ground-truth segments).

The **baseline** scorer receives only the local per-frame BiLSTM hidden state $\mathbf{h}_t \in \mathbb{R}^{512}$:

$$z_t^{\text{base}} = \mathbf{w}_o^\top \mathbf{h}_t + b_o, \tag{9}$$

trained with the masked binary cross-entropy loss $\mathcal{L}_{\text{BCE}}$ (Eq. 2), AdamW with weight decay 10^{-5}, and a StepLR scheduler that halves the learning rate every 15 epochs regardless of validation-loss behavior.

B. Modification 1: Attention-Context Fusion in the Scorer

1) Motivation and Theoretical Insight: Equation (9) scores frame t using *only local context*: the BiLSTM hidden state $\mathbf{h}_t$ encodes the neighborhood around t, but the scorer has no explicit access to the *global distribution* of content across the entire clip. This is a fundamental limitation: the same motion or speech pattern may be highly salient in a short clip but unremarkable in a two-hour documentary. Without a global reference, the scorer cannot adapt its threshold to the clip's intrinsic content distribution.

Attention-augmented encoder–decoder architectures for video summarization [13], [16] address this by explicitly

routing a globally pooled context vector into the scoring layer. Intuitively, the global context $\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t$ encodes *what the clip is about on average*, weighted by attention scores; when a frame's local state $\mathbf{h}_t$ is concatenated with $\mathbf{c}$, the scorer can learn rules of the form: "frame t is salient *because it is unusual relative to the clip average*," or conversely, "it is unimportant *because it merely repeats the dominant pattern.*" This comparative reasoning is precisely the judgment a human editor applies when selecting highlights.

2) Mathematical Formulation: Let $\mathbf{c} \in \mathbb{R}^{512}$ denote the Temporal Attention context vector (identical for all t but encoding the full sequence):

$$\mathbf{u}_t = \big[\mathbf{h}_t \,;\, \mathbf{c}\big] \in \mathbb{R}^{1024}, \qquad z_t^{\text{opt}} = \mathbf{W}_o\, \mathbf{u}_t + b_o, \tag{10}$$

where $\mathbf{W}_o \in \mathbb{R}^{1\times 1024}$. The concatenated vector $\mathbf{u}_t$ allows the first linear layer of the scorer to model multiplicative interactions between local and global representations—for instance, suppressing a frame whose local state closely matches $\mathbf{c}$ (redundant content), or amplifying a frame that deviates from $\mathbf{c}$ in semantically relevant dimensions.

3) Implementation Consequence: Because the scorer's first weight matrix changes shape from $\mathbb{R}^{1\times 512}$ (baseline) to $\mathbb{R}^{1\times 1024}$ (optimized), the two checkpoints are **weight-incompatible at the scorer layer**. Loading a baseline checkpoint into the optimized architecture (or vice versa) produces a shape mismatch and meaningless output scores. Full retraining with the correct `use_context_fusion` flag is mandatory before evaluation.

C. Modification 2: Hybrid Quality–Diversity Keyshot Selection

1) Motivation and Theoretical Insight: The baseline post-processing selects the top-k shots by *pure greedy score ranking*: shots are sorted by their smoothed per-frame score and the highest-scoring ones are selected until the duration budget $r \cdot |V|$ is exhausted. This strategy maximizes per-shot saliency but has a well-known failure mode: if the model assigns high scores to a *compact temporal cluster* (e.g., the climax of a lecture near its midpoint), the entire budget may be consumed by that cluster, producing a summary that covers only a fraction of the clip's actual time span. The resulting summary has high precision (every selected frame is indeed salient) but low recall (large portions of the video are never represented), which depresses F-score and, more critically, temporal overlap.

This trade-off between *saliency* and *temporal coverage* is a central theme in submodular optimization for summarization [44], [45]: a pure quality objective is concave and benefits from diminishing returns only if the coverage term is explicitly included. The hybrid strategy below approximates a coverage-aware submodular selection without the computational cost of full submodular maximization.

2) Mathematical Formulation: After shot-level score smoothing, the total frame budget is $k_{\text{tot}} \approx \lceil r \cdot L \rceil$. Define a quality fraction $\rho \in (0, 1)$ (default $\rho = 0.7$):

$$k_q = \text{round}(\rho\, k_{\text{tot}}), \qquad k_d = k_{\text{tot}} - k_q. \tag{11}$$

Phase 1 (quality): Select the k_q highest-scoring shots by descending model score. Let $\mathcal{S}_q$ denote this set.

Phase 2 (diversity): Partition the video timeline into B equal temporal bins. For each bin $b \notin \{\text{bins already covered by } \mathcal{S}_q\}$, add the highest-scoring unselected shot from b until k_d shots are added, forming $\mathcal{S}_d$.

The final selection is $\mathcal{S} = \mathcal{S}_q \cup \mathcal{S}_d$. This ensures at least $B - |\mathcal{S}_q|$ distinct temporal regions are covered, which reduces the probability that the ground-truth annotation segments (which are distributed across the timeline) fall entirely into the unselected portion of the video.

3) Practical Considerations: Setting $\rho = 1$ recovers pure greedy top-k; setting $\rho = 0$ produces uniform forced coverage regardless of quality. Intermediate values of ρ provide a smooth interpolation. The choice $\rho = 0.7$ was motivated by the empirical observation that TVSum annotation density is non-uniform and biased toward the first and last quarters of each video; allocating 30% of the budget to coverage prevents systematic under-representation of the video's tails. For datasets with more uniform annotations (e.g., SumMe), a higher ρ closer to 0.8–0.9 may be preferable to avoid including low-salience filler from quiet bins.

D. Modification 3: Auxiliary Segment-Coverage Loss

1) Motivation and Theoretical Insight: The baseline training objective (masked BCE, Eq. 2) is a purely pointwise loss: it evaluates each frame independently and does not penalize *structural* properties of the predicted score distribution, such as temporal concentration. A model trained with pure BCE can reach a local optimum where it assigns high probabilities to a single short segment (e.g., the loudest speech burst) and near-zero probabilities everywhere else, because this behavior minimizes BCE on a dataset where ground-truth positives are clustered. The post-processing pipeline can partially compensate for this via the diversity phase of Modification 2, but it is more principled—and empirically more effective—to encode the coverage requirement directly into the loss.

Multi-objective training with auxiliary supervision signals is a standard technique for injecting structural inductive biases that the primary loss cannot express [46]. The auxiliary diversity penalty below acts as a *list-level* regularizer: instead of evaluating each frame's score in isolation, it evaluates the *distribution of peak scores across temporal segments*, encouraging the model to spread its predictive confidence across the video.

2) Mathematical Formulation: Partition the sequence of T frames into K contiguous non-overlapping segments $\{S_j\}_{j=1}^{K}$ of equal length (controlled by `n_segments`, default $K = 10$). Define the *segment peak score* as the maximum predicted probability within each segment:

$$m_j = \max_{t \in S_j} \sigma(z_t), \qquad j = 1, \ldots, K. \tag{12}$$

The **auxiliary diversity loss** penalizes high variance in the segment peak scores:

$$\mathcal{L}_{\text{div}} = \text{Var}(m_1, \ldots, m_K) = \frac{1}{K}\sum_{j=1}^{K}(m_j - \bar{m})^2, \quad \bar{m} = \frac{1}{K}\sum_j m_j. \tag{13}$$

The **total loss** is a weighted combination:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda\,\mathcal{L}_{\text{div}}, \tag{14}$$

where $\lambda = \texttt{diversity_weight} \geq 0$.

3) Behavioral Analysis: To understand the gradient signal from $\mathcal{L}_{\text{div}}$, consider two extremes. If all segment peaks are equal ($m_j = c\ \forall j$), then $\mathcal{L}_{\text{div}} = 0$: the model has already distributed its predictive confidence uniformly, and no penalty is applied. If one segment has $m_j = 1$ and all others have $m_j \approx 0$, then $\mathcal{L}_{\text{div}} \approx (1-1/K)^2/K \cdot K = (K-1)/K^2$, which is the maximum variance for a binary peak distribution. The gradient of $\mathcal{L}_{\text{div}}$ with respect to z_t for $t \in \arg\max_{t \in S_j} \sigma(z_t)$ is:

$$\frac{\partial \mathcal{L}_{\text{div}}}{\partial z_t} = \frac{2\lambda}{K}(m_j - \bar{m})\, m_j(1 - m_j), \tag{15}$$

which is *positive* for segments with above-average peaks (pushing their maximum score down toward the mean) and *negative* for below-average segments (pushing their maximum score up). The net effect is to "flatten" the distribution of segment confidence peaks—the model is incentivized to find locally high-scoring frames in each temporal region rather than converging on a single dominant one.

4) Hyperparameter Sensitivity: Setting $\lambda = 0$ recovers the baseline. Excessively large λ causes the diversity loss to dominate, degrading the primary ranking quality: the model may assign nearly uniform scores to all frames, which collapses the discrimination between salient and non-salient content. In practice, $\lambda \in [0.01, 0.1]$ provides a useful range on the SumMe/TVSum benchmarks; the optimal value should be tuned on a held-out validation set.

The choice of K also matters. A small K (e.g., $K = 3$) applies coarse coverage pressure—sufficient to prevent severe temporal concentration but insensitive to within-segment clustering. A large K close to the number of shots applies fine-grained coverage pressure but makes the variance statistic noisy when segments contain very few frames. The default $K = 10$ corresponds to roughly one segment per minute of footage at a 2 fps sampling rate with a 960-frame cap, which aligns well with the average TVSum video length of ≈5–7 min.

E. Modification 4: Regularization and Adaptive Learning-Rate Scheduling

1) Motivation and Theoretical Insight: The two preceding modifications (context fusion and auxiliary loss) jointly increase the effective model capacity: the scorer now operates on a 1024-dimensional input (vs. 512 in the baseline), and the gradient signal includes an additional structural term. Without corresponding changes to the regularization and learning-rate schedule, the optimized model faces an increased risk of overfitting—particularly on SumMe, which contains fewer than 25 training videos in the standard split.

Furthermore, the baseline StepLR scheduler decays the learning rate by a fixed factor every 15 epochs, *regardless of whether the model has converged or is still improving*. In the early training phase this can cause premature learning-rate decay; in the late phase, it can delay decay when the model is oscillating near a flat region. Both behaviors are avoidable with an *adaptive* scheduler that monitors validation loss directly [43].

2) Increased Weight Decay (AdamW): AdamW [41] decouples weight decay from the adaptive gradient scaling of Adam, so the ℓ_2 regularization term acts directly on the weight magnitudes rather than being modulated by per-parameter learning-rate estimates. In the baseline, the weight decay coefficient is $\lambda_{\text{wd}} = 10^{-5}$. In the optimized variant it is increased to $\lambda_{\text{wd}} = 10^{-4}$.

The motivation is two-fold. First, the wider 1024-dimensional scorer has 2× as many parameters at the output layer, which increases the raw capacity for overfitting. Second, the diversity loss introduces an additional gradient signal that can cause weight magnitudes to grow rapidly in the early training epochs before the BCE loss stabilizes; a stronger weight-decay coefficient counteracts this growth. The tenfold increase is conservative: $\lambda_{\text{wd}} = 10^{-4}$ is still well within the "light regularization" regime for an architecture of this size, and is consistent with the default recommendation for transformer-based models [42].

3) ReduceLROnPlateau Scheduler: The baseline StepLR applies a fixed decay:

$$\eta_k = \eta_0 \cdot \gamma^{\lfloor k/\text{step} \rfloor}, \quad \gamma = 0.5, \quad \text{step} = 15. \tag{16}$$

The optimized variant replaces this with `ReduceLROnPlateau` (PyTorch [43]), which monitors the validation loss $\mathcal{L}^{\text{val}}$ and applies decay only when no improvement is observed for `patience` consecutive epochs:

$$\eta_{k+1} = \begin{cases} \eta_k \cdot \text{factor} & \text{if } \mathcal{L}_k^{\text{val}} \geq \min_{j \leq k-\text{patience}} \mathcal{L}_j^{\text{val}} \\ \eta_k & \text{otherwise.} \end{cases} \tag{17}$$

The default parameters are `factor` $= 0.5$ and `patience` $= 5$. This formulation has two concrete advantages over StepLR in the context of this work. First, when the auxiliary diversity loss $\mathcal{L}_{\text{div}}$ introduces a noisy gradient that causes transient spikes in validation loss, the plateau scheduler will not incorrectly trigger on a single bad epoch (patience provides a buffer). Second, the scheduler remains at a high learning rate for as long as the model is making consistent progress, allowing longer useful training on smaller datasets where the number of gradient steps is the primary bottleneck.

4) Interaction with Early Stopping: Both the plateau scheduler and early stopping use validation loss as the monitoring signal. To avoid double-counting the patience period, the early stopping counter is reset each time the scheduler triggers a decay. In practice this means the model receives at least

`patience`× `factor` learning-rate epochs after each reduction before training is allowed to terminate, which prevents premature stopping immediately after a decay step when the model has not yet had time to adapt to the new learning rate.

F. Quantitative Results and Interpretation

Table XI reports a representative single-seed measurement on the held-out test split.

TABLE XI: Baseline vs. Optimized Variant: Test-Set Metrics

Metric	Baseline	Optimized	Δ
Precision	0.4767	0.5293	+0.0526
Recall	0.4767	0.5293	+0.0526
F-score	0.4767	0.5293	+0.0526
Temporal overlap	0.2347	0.3709	+0.1363

1) F-Score Improvement (+5.3 pp): The +5.3 pp gain in F-score reflects improved *per-frame discrimination*: the optimized model selects frames that more consistently match the binarized ground-truth labels. The primary driver is Modification 1 (context fusion): by conditioning the scorer on the global context vector $\mathbf{c}$, the model learns to rank frames *relative to the clip's own distribution* rather than applying a fixed threshold. This is equivalent to normalizing the predicted scores by a clip-level reference, which reduces false positives in visually monotonous clips and false negatives in clips with uniformly rich content.

The auxiliary diversity loss (Modification 3) contributes to recall specifically: by penalizing score concentration, it prevents the model from producing a small number of very high-confidence predictions that exhaust the top-k budget, which under the top-k symmetric evaluation protocol (Section VIII) would lower recall without a corresponding gain in precision.

2) Temporal Overlap Improvement (+13.6 pp): The +13.6 pp gain in temporal overlap is substantially larger than the F-score gain and reflects *structural improvement in summary layout* rather than per-frame accuracy. Frame-level F-score and temporal overlap are correlated but measure different things: F-score rewards selecting the individually most important frames, while temporal overlap rewards selecting frames from the same *temporal regions* as the ground-truth—even if the exact frame boundaries differ by a few seconds. A summary that consists of short scattered clips spread across the full video will have higher temporal overlap than one that concentrates all clips in the video's highest-salience minute.

The hybrid keyshot selection (Modification 2) is the principal driver of the overlap gain. Phase 2 of the selection algorithm explicitly allocates budget to under-represented temporal bins, directly increasing the probability that selected segments intersect ground-truth intervals distributed across the timeline. The diversity loss (Modification 3) provides complementary pressure at training time, so the score distribution going into the selection algorithm is already more temporally spread, reducing the work left for Phase 2 to do.

3) Precision–Recall Symmetry: As discussed in Section VIII, the top-k binarization protocol equalizes the number of positive predictions and positive ground-truth frames, causing Precision $\approx$ Recall in the aggregate metric. Both equal 0.5293 in the optimized variant, confirming that this symmetry holds at the new operating point as well. The per-dataset F-scores (SumMe, TVSum) are unaffected by this symmetry because they are computed from separate subsets.

4) Expected vs. Additive Gains: The four modifications are applied jointly, so Table XI reports their *combined* effect. Formal attribution of each component's individual contribution requires controlled single-modification ablation experiments with matched seeds and splits, which are left for future work. Based on the theoretical motivation above, the expected ordering of individual contributions to temporal overlap is: Hybrid Selection > Diversity Loss > Context Fusion > Regularization/Scheduling; and for F-score: Context Fusion > Diversity Loss > Hybrid Selection > Regularization/Scheduling. The regularization changes are expected to primarily affect *generalization variance* across seeds rather than the mean performance at a single seed, so their contribution to Table XI may be underestimated by a single-run evaluation.

5) Dataset-Specific Caveats: The improvements in Table XI are reported on the combined SumMe/TVSum test split. As discussed in Section VIII, the two datasets have substantially different annotation styles, video lengths, and speech characteristics. Context fusion is expected to benefit TVSum (semantically rich, long videos where global context is more informative) more than SumMe (short user-generated clips where global context adds little additional information beyond what a local BiLSTM already captures). The hybrid selection strategy may benefit SumMe more, since SumMe's shorter videos have more uniform annotation density. Disaggregating the results by dataset is an important direction for future work.

G. Summary of Modifications

TABLE XII: Summary of Optimized Variant Modifications and Their Primary Effects

Modification	Primary Bottleneck Addressed	Expected Main Gain
Context fusion (Eq. 10)	Local-only scoring; no global reference	F-score (discrimination)
Hybrid keyshot selection (Eq. 11)	Temporal concentration of top-k	Temporal overlap (coverage)
Auxiliary diversity loss (Eq. 14)	BCE promotes score concentration	Overlap + Recall
Increased λ_{wd} + ReduceLROnPlateau (Eq. 17)	Overfitting; premature / delayed LR decay	Generalization stability

All four modifications target *different* aspects of the training and inference pipeline and are largely orthogonal in their mechanism: context fusion changes the *scoring function*; hybrid selection changes the *post-processing strategy*; the

diversity loss changes the *training objective*; and the regularization/scheduling changes change the *optimization trajectory*. Their complementary nature is the main reason the joint system outperforms the baseline across all four reported metrics simultaneously.

XI. Conclusion

This report presented a **multimodal BiLSTM with Temporal Attention** using **2432-dimensional** feature vectors, trained with supervised learning at summary ratio r=0.35. Ablation experiments confirm that the full model achieves a TVSum F-score of 60.1%, competitive with published LSTM-based baselines, while SumMe results highlight dataset sensitivity to audio-branch noise and training budget. The per-component analysis shows that bidirectionality and Temporal Attention are most beneficial for speech-rich content, while the audio branch's contribution depends strongly on transcript quality and video domain.

The **optimized variant** (Section X) further improves F-score by $+5.3$ pp and temporal overlap by $+13.6$ pp over the baseline through attention-context fusion, hybrid keyshot selection, an auxiliary coverage loss, and adaptive regularization.

The **VideoIntel** evaluation demonstrates that **CLIP semantic search** on summaries suffers **negligible degradation**, while **Q&A quality** depends critically on transcript density and topical accuracy—reflecting video summarization as **lossy compression** and underlining the need for **downstream task validation** before deployment.

Appendix

This appendix reports a **downstream experiment** independent of the BiLSTM training on SumMe/TVSum. The goal is to assess whether videos produced by the summarization pipeline retain sufficient behavioral signal to train a simple event detector, compared to the original full-length videos. The task is fully controlled: identical feature type, identical classifier (random forest), and two training conditions differing only in video source (original vs. summary) and the time-axis label mapping for the summary branch.

A. Motivation and Hypothesis

Condition A trains on full-length videos; **Condition B** trains on the corresponding summaries. If the retained shots cover the high-motion regions associated with events (e.g., collisions, hard braking), Condition B may achieve lower or comparable temporal MAE owing to shorter sequences and reduced noise. Conversely, if the summarizer discards or misaligns the event-containing segments, Condition B will underperform.

Event timestamps for Condition B are obtained by accumulating the durations of retained segments in order: if a ground-truth event timestamp falls within a retained segment, it is mapped to the corresponding position on the summary timeline; if it falls in a discarded region, the sample may be excluded depending on the annotation convention.

B. Task Definition

Input. A single video file; the system samples frames at a fixed rate (e.g., 4 fps) and computes a fixed-dimensional statistical feature vector without any deep network at this stage.

Outputs. (1) *Binary classification*: event present / absent; (2) *Temporal regression*: predicted event timestamp $\hat{t}$ (seconds), clipped to $[0, T]$ where T is the video duration.

Feature vector (per video). Duration, sampled frame count, mean and standard deviation of inter-frame motion magnitude (computed on downsampled grayscale frames via OpenCV [50]), peak motion value, brightness statistics, and the temporal position of the motion peak.

C. Pilot Dataset

TABLE XIII: Pilot Dataset Split for Event Detection

Split	Videos
Training	31
Validation	8
Test	6
Total	**45**

All labels were assigned manually. **Limitation:** every sample in the current splits carries a positive event label; accordingly, accuracy and F1 are uninformative about negative-class discrimination. The central comparative metric is **temporal MAE** between the two conditions.

D. Training Procedure

- **Classifier:** random forest [49] with class-balanced sample weights (300 trees).
- **Temporal regressor:** a separate random forest (200 trees) trained only on positive-class samples with valid timestamps.

Both conditions share identical hyperparameters and feature extraction; only the video source and remapped labels differ.

E. Evaluation Criteria

Accuracy and F1 are reported for completeness but are not discriminative under an all-positive label distribution. The primary metric is:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|t_{\mathrm{gt},i} - \hat{t}_i\right| \quad \text{(seconds)}. \tag{18}$$

F. Results

TABLE XIV: Event Detection Results: Condition A (Original) vs. B (Summary)

Split	n	Acc / F1 (A / B)		MAE in s (A / B)	
Training	31	1.00 / 1.00	1.00 / 1.00	0.305	0.470
Validation	8	1.00 / 1.00	1.00 / 1.00	1.622	0.292
Test	6	1.00 / 1.00	1.00 / 1.00	0.549	1.269

On the **validation** split, Condition B achieves lower MAE; on the **test** split, Condition A performs better. With only 8 and 6 videos respectively, this reversal does not permit statistically robust conclusions. The results suggest that summary-based features can be competitive when the retained shots align with event regions, but the small scale prevents generalization.

G. Discussion and Future Directions

Summary videos may concentrate motion statistics around retained segments, benefiting temporal regression on splits where those segments coincide with event regions, while degrading on splits where the summarizer misaligned the selection. Scaling the dataset, adding negative-class samples, and adopting deep learned features (e.g., I3D [37] or TimeSformer [35]) are natural extensions toward a rigorous evaluation.

Numerical results for all ablation variants—visual-only BiLSTM, full multimodal BiLSTM, BiLSTM without Temporal Attention, and unidirectional UniLSTM—are reported in Table III of Section VIII. All variants share the same evaluation protocol (fixed split, fixed summary ratio $r{=}0.35$, identical post-processing), enabling direct attribution of performance differences to individual architectural choices.

The incremental contributions of architectural components vary by dataset. On **TVSum**, the full multimodal model achieves the highest F-score (60.1%), confirming the benefit of the audio-language branch for speech-rich content (+0.4 pp over visual-only). On **SumMe**, simpler variants outperform the full model in this single-seed run, consistent with the training-budget and audio-noise sensitivity discussed in Section VIII. These findings align with the broader literature on sequential summarization models [11], [16]: architectural gains are dataset-dependent and require multi-seed evaluation for robust attribution.

Extensions to new content domains or additional downstream tasks (content classification, captioning, retrieval on summarized video) are outside the scope of this report but are motivated by the analysis in Section X and Appendix A.

References

[1] M. Gygli, H. Grabner, H. Riemenschneider, and L. Van Gool, "Creating summaries from user videos," in *ECCV*, vol. 8695, 2014, pp. 505–520.

[2] Y. Song, J. Vallmitjana, A. Stent, and A. Jaimes, "TVSum: Summarizing web videos using titles," in *CVPR*, 2015, pp. 5179–5187.

[3] E. Apostolidis, E. Adamantidou, A. I. Metsai, V. Mezaris, and I. Patras, "Video summarization using deep neural networks: A survey," *Proc. IEEE*, vol. 109, no. 11, pp. 1838–1863, 2021.

[4] H. B. Haq, M. Asif, M. B. Ahmad, R. Ashraf, and T. Mahmood, "Video summarization techniques: A review," *Math. Probl. Eng.*, vol. 2021, pp. 1–17, 2021.

[5] A. G. Money and H. Agius, "A survey on video summarization techniques," *Int. J. Comput. Appl.*, vol. 118, no. 11, pp. 25–31, 2015.

[6] P. Kadam *et al.*, "Recent challenges and opportunities in video summarization with machine learning algorithms," *IEEE Access*, vol. 10, pp. 122762–122785, 2022.

[7] S. Hu, Z. Liu, J. Liu, and Z. Guo, "Video summarization based on feature fusion and data augmentation," *Computers*, vol. 12, no. 9, p. 186, 2023.

[8] V. Tiwari and C. Bhatnagar, "A survey of recent work on video summarization: Approaches and techniques," *Multimed. Tools Appl.*, vol. 80, no. 18, pp. 27187–27221, 2021.

[9] P. Meena, H. Kumar, and S. K. Yadav, "A review on video summarization techniques," *Eng. Appl. Artif. Intell.*, vol. 118, p. 105667, 2023.

[10] Q. Yu, Z. Wang, G. Wei, and H. Yu, "Deep learning for video summarization: Systematic review, challenges and opportunities," *IEEE/CAA J. Autom. Sinica*, vol. 13, no. 1, pp. 21–42, Jan. 2026.

[11] K. Zhang, W.-L. Chao, F. Sha, and K. Grauman, "Video Summarization with Long Short-Term Memory (vsLSTM, dppLSTM)," in *ECCV*, 2016.

[12] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] H. Ji, M. Xu, and J. Yang, "Video Summarization with Attention-Based Encoder-Decoder Networks," *arXiv:1708.09545*, 2017.

[14] Z. Ji, K. Xiong, Y. Pang, and X. Li, "Video summarization with attention-based encoder–decoder networks," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1709–1717, 2019.

[15] J. Lin, S.-H. Zhong, and A. Fares, "Deep hierarchical LSTM networks with attention for video summarization," *Computers & Electrical Engineering*, vol. 97, p. 107618, 2022.

[16] J. Fajtl, H. S. Sokeh, V. Argyriou, D. Monekosso, and P. Remagnino, "Summarizing Videos with Attention (VASNet)," in *ACCV*, 2018.

[17] A. Vaswani *et al.*, "Attention Is All You Need," in *Adv. Neural Inf. Process. Syst. (NIPS)*, 2017.

[18] L. Lan *et al.*, "FullTransNet: Full Transformer with Local-Global Attention for Video Summarization," *arXiv:2501.00882*, 2025.

[19] E. Apostolidis, K. Mekkas, A. I. Patras, and I. Kompatsiaris, "PGL-SUM: A Novel Pointer-Generator Network for Extractive Video Summarization," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, 2021.

[20] S. A. Ghauri, F. S. Khan, S. W. Zamir, J. Hayat, and M. Shah, "MSVA: Multi-Stage Aggregated Transformer for Video Summarization," *arXiv:2104.11530*, 2021.

[21] Y. Zhu *et al.*, "MHSCNet: Multi-Head Self-Attention Based Convolutional Network for Video Summarization," *arXiv:2204.08352*, 2022.

[22] B. Mahasseni, M. Lam, and S. Todorovic, "Unsupervised Video Summarization with Adversarial LSTM Networks (SUM-GAN)," in *CVPR*, 2017.

[23] K. Zhang, K. Grauman, and F. Sha, "Retrospective Encoders for Video Summarization (Cycle-SUM)," in *AAAI*, 2019.

[24] E. Apostolidis, E. Adamantidou, V. Mezaris, and I. Patras, "Self-Attention Based Generative Adversarial Networks for Unsupervised Video Summarization (SUM-GAN-AED)," *arXiv:2307.08145*, 2023.

[25] K. Zhou, Y. Qiao, and T. Xiang, "Deep Reinforcement Learning for Unsupervised Video Summarization with Diversity-Representativeness Reward," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, 2018.

[26] W. Zhu, Y. Han, J. Lu, and J. Zhou, "Relational Reasoning Over Spatial-Temporal Graphs for Video Summarization," *IEEE Trans. Image Process.*, vol. 31, pp. 3017–3031, 2022.

[27] T. Psallidas, P. Koromilas, T. Giannakopoulos, and E. Spyrou, "Multimodal summarization of user-generated videos," *Appl. Sci.*, vol. 11, no. 11, p. 5260, 2021.

[28] B. He, J. Wang, J. Qiu, T. Bui, A. Shrivastava, and Z. Wang, "Align and Attend: Multimodal Summarization with Dual Contrastive Losses," in *CVPR*, 2023.

[29] M. Narasimhan, A. Rohrbach, and T. Darrell, "CLIP-It! Language-Guided Video Summarization," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, pp. 13988–14000, 2021.

[30] H. Li, Q. Ke, M. Gong, and T. Drummond, "Progressive Video Summarization via Multimodal Self-Supervised Learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2023, pp. 5584–5593.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016.

[32] A. Radford *et al.*, "Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)," in *ICML*, 2023.

[33] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *EMNLP*, 2019.

[34] A. Radford *et al.*, "Learning Transferable Visual Models From Natural Language Supervision (CLIP)," in *ICML*, 2021.

[35] G. Bertasius, H. Wang, and L. Torresani, "Is Space-Time Attention All You Need for Video Understanding? (TimeSformer)," in *ICML*, 2021.

[36] Z. Liu *et al.*, "Video Swin Transformer," in *CVPR*, 2022.

[37] J. Carreira and A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset (I3D)," in *CVPR*, 2017.

[38] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL*, 2019.

[39] W. Zhu, J. Lu, J. Li, and J. Zhou, "DSNet: A Flexible Detect-to-Summarize Network for Video Summarization," *IEEE Trans. Image Process.*, 2021.

[40] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *ICLR*, 2015.

[41] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization (AdamW)," in *ICLR*, 2019.

[42] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[43] PyTorch Developers, "`ReduceLROnPlateau`," in *PyTorch Documentation (Stable)*, https://pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.ReduceLROnPlateau.html.

[44] M. Gygli, H. Grabner, and L. Van Gool, "Video Summarization by Learning Submodular Mixtures of Objectives," in *CVPR*, 2015.

[45] H. Lin and J. Bilmes, "A Class of Submodular Functions for Document Summarization," in *Proc. NAACL*, 2011.

[46] R. Caruana, "Multitask Learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[47] FFmpeg Developers, *FFmpeg: A Complete, Cross-Platform Solution to Record, Convert and Stream Audio and Video*, https://ffmpeg.org.

[48] Ollama Developers, *Ollama: Run Large Language Models Locally*, https://ollama.com.

[49] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[50] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.

[51] B. Zhao, X. Li, and X. Lu, "HSA-RNN: Hierarchical Structure-Adaptive RNN for Video Summarization," in *CVPR*, 2018.

[52] Y. Jung, D. Cho, D. Kim, S. Woo, and I.-S. Kweon, "Discriminative Feature Learning for Unsupervised Video Summarization," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, pp. 8537–8544, 2019.

[53] K. Zhou, Y. Qiao, and T. Xiang, "Deep reinforcement learning for unsupervised video summarization with diversity-representativeness reward," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, 2018.

[54] G. Wang, X. Wu, and J. Yan, "Progressive reinforcement learning for video summarization," *Information Sciences*, vol. 655, p. 119888, 2024.

[55] T. Liu *et al.*, "Video Summarization through Reinforcement Learning with a 3D Spatio-Temporal U-Net," *IEEE Trans. Image Process.*, vol. 31, 2022.

[56] X. Shang, Z. Yuan, A. Wang, and C. Wang, "Multimodal Video Summarization via Time-Aware Transformers," in *Proc. ACM Int. Conf. Multimedia (ACM MM)*, 2021, pp. 1756–1765.

[57] A. Miech *et al.*, "HowTo100M: Learning a Text-Video Embedding by Watching Hundred Million Narrated Video Clips," in *ICCV*, 2019.

[58] A. Dosovitskiy *et al.*, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT)," in *ICLR*, 2021.

[59] R. Zhong *et al.*, "Semantic Representation and Attention Alignment for Graph Information Bottleneck in Video Summarization," *IEEE Trans. Image Process.*, 2023.

[60] H. Khan, T. Hussain, S. U. Khan, Z. A. Khan, and S. W. Baik, "Deep multi-scale pyramidal features network for supervised video summarization," *Expert Syst. Appl.*, vol. 237, p. 121288, 2024.